```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import MinMaxScaler
from sklearn.metrics import mean_absolute_error, mean_squared_error

from sklearn.ensemble import RandomForestRegressor,
GradientBoostingRegressor
from sklearn.linear_model import LinearRegression
```

Step 3: Assign Column Names

```
columns = ['engine_id','cycle','op1','op2','op3'] + ['sensor_'+str(i)
for i in range(1,22)]

print(columns)
```

```
['engine_id', 'cycle', 'op1', 'op2', 'op3', 'sensor_1', 'sensor_2',
'sensor_3', 'sensor_4', 'sensor_5', 'sensor_6', 'sensor_7',
'sensor_8', 'sensor_9', 'sensor_10', 'sensor_11', 'sensor_12',
'sensor_13', 'sensor_14', 'sensor_15', 'sensor_16', 'sensor_17',
'sensor_18', 'sensor_19', 'sensor_20', 'sensor_21']
```

```
train = pd.read_csv("train_FD001.txt", sep=r"\s+", header=None)
train.columns = columns

train.head()
```

```
   engine_id  cycle     op1     op2    op3  sensor_1  sensor_2
sensor_3  \
0          1      1 -0.0007 -0.0004  100.0    518.67    641.82
1589.70
1          1      2  0.0019 -0.0003  100.0    518.67    642.15
1591.82
2          1      3 -0.0043  0.0003  100.0    518.67    642.35
1587.99
3          1      4  0.0007  0.0000  100.0    518.67    642.35
1582.79
4          1      5 -0.0019 -0.0002  100.0    518.67    642.37
1582.85

   sensor_4  sensor_5  ...  sensor_12  sensor_13  sensor_14  sensor_15
\
0   1400.60     14.62  ...     521.66    2388.02    8138.62     8.4195

1   1403.14     14.62  ...     522.28    2388.07    8131.49     8.4318
```

```
2   1404.20     14.62  ...     522.42    2388.03    8133.23     8.4178

3   1401.87     14.62  ...     522.86    2388.08    8133.83     8.3682

4   1406.22     14.62  ...     522.19    2388.04    8133.80     8.4294


   sensor_16  sensor_17  sensor_18  sensor_19  sensor_20  sensor_21
0       0.03        392       2388      100.0      39.06    23.4190
1       0.03        392       2388      100.0      39.00    23.4236
2       0.03        390       2388      100.0      38.95    23.3442
3       0.03        392       2388      100.0      38.88    23.3739
4       0.03        393       2388      100.0      38.90    23.4044

[5 rows x 26 columns]
```

```
train.shape
```

```
(20631, 26)
```

```
max_cycle = train.groupby('engine_id')['cycle'].max()

train = train.merge(max_cycle.to_frame(name='max_cycle'),
on='engine_id')

train['RUL'] = train['max_cycle'] - train['cycle']

train.head()
```

```
   engine_id  cycle     op1     op2    op3  sensor_1  sensor_2
sensor_3  \
0          1      1 -0.0007 -0.0004  100.0    518.67    641.82
1589.70
1          1      2  0.0019 -0.0003  100.0    518.67    642.15
1591.82
2          1      3 -0.0043  0.0003  100.0    518.67    642.35
1587.99
3          1      4  0.0007  0.0000  100.0    518.67    642.35
1582.79
4          1      5 -0.0019 -0.0002  100.0    518.67    642.37
1582.85

   sensor_4  sensor_5  ...  sensor_14  sensor_15  sensor_16  sensor_17
\
0   1400.60     14.62  ...    8138.62     8.4195       0.03        392

1   1403.14     14.62  ...    8131.49     8.4318       0.03        392

2   1404.20     14.62  ...    8133.23     8.4178       0.03        390
```

```
3   1401.87     14.62  ...    8133.83     8.3682       0.03        392

4   1406.22     14.62  ...    8133.80     8.4294       0.03        393


   sensor_18  sensor_19  sensor_20  sensor_21  max_cycle  RUL
0       2388      100.0      39.06    23.4190        192  191
1       2388      100.0      39.00    23.4236        192  190
2       2388      100.0      38.95    23.3442        192  189
3       2388      100.0      38.88    23.3739        192  188
4       2388      100.0      38.90    23.4044        192  187

[5 rows x 28 columns]
```

```
train.shape
```

```
(20631, 28)
```

Step 5: Data Cleaning

```
unique_counts =train.nunique()
constant_columns = unique_counts[unique_counts == 1].index.tolist()

train.drop(constant_columns, axis=1, inplace=True)

unique_counts
#constant_columns
```

```
engine_id     100
cycle         362
op1           158
op2            13
op3             1
sensor_1        1
sensor_2      310
sensor_3     3012
sensor_4     4051
sensor_5        1
sensor_6        2
sensor_7      513
sensor_8       53
sensor_9     6403
sensor_10       1
sensor_11     159
sensor_12     427
sensor_13      56
sensor_14    6078
sensor_15    1918
sensor_16       1
sensor_17      13
```

```
sensor_18       1
sensor_19       1
sensor_20     120
sensor_21    4745
max_cycle      66
RUL           362
dtype: int64
```

```
constant_columns
```

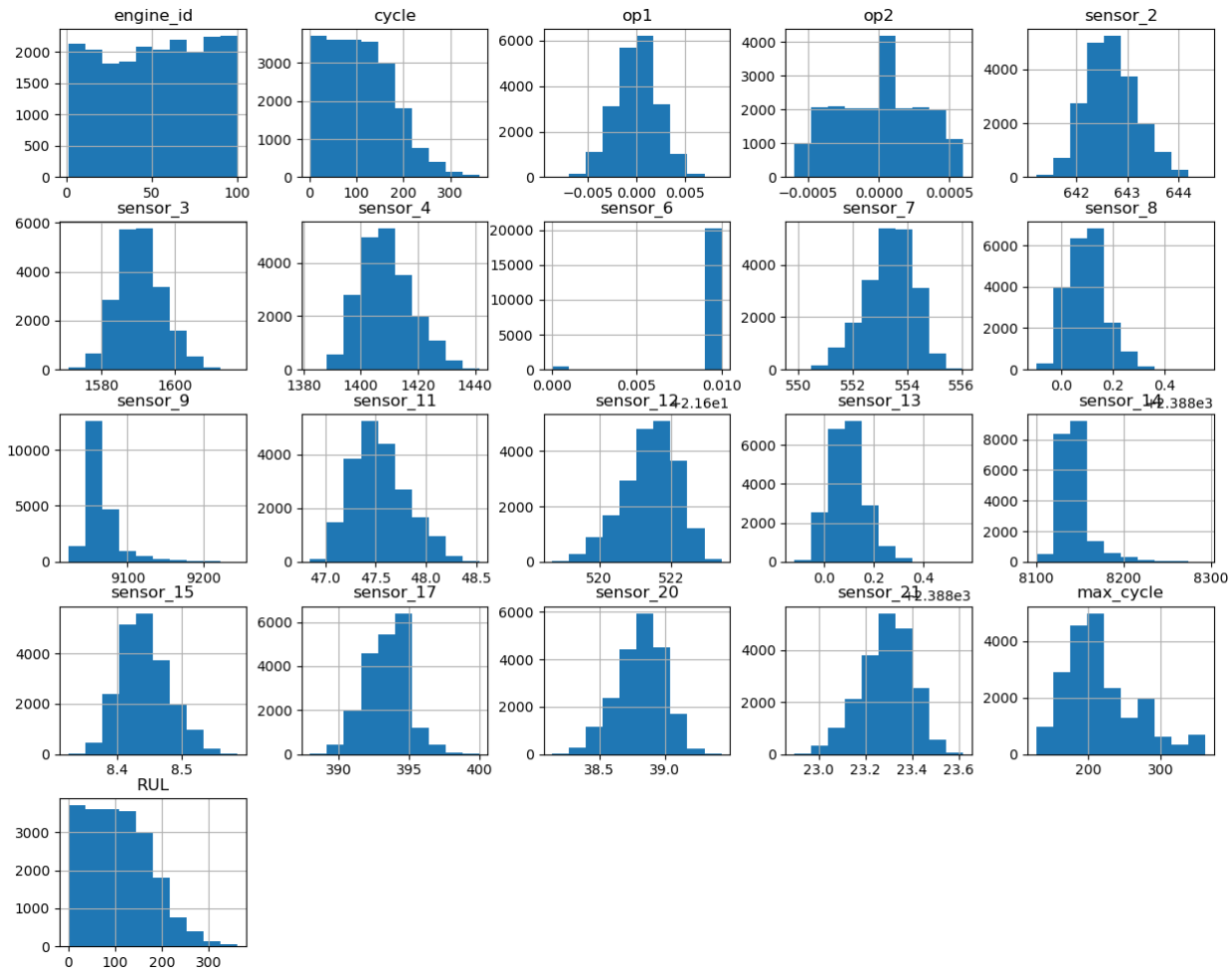
```
['op3',
 'sensor_1',
 'sensor_5',
 'sensor_10',
 'sensor_16',
 'sensor_18',
 'sensor_19']
```

```
train.shape
```

```
(20631, 21)
```

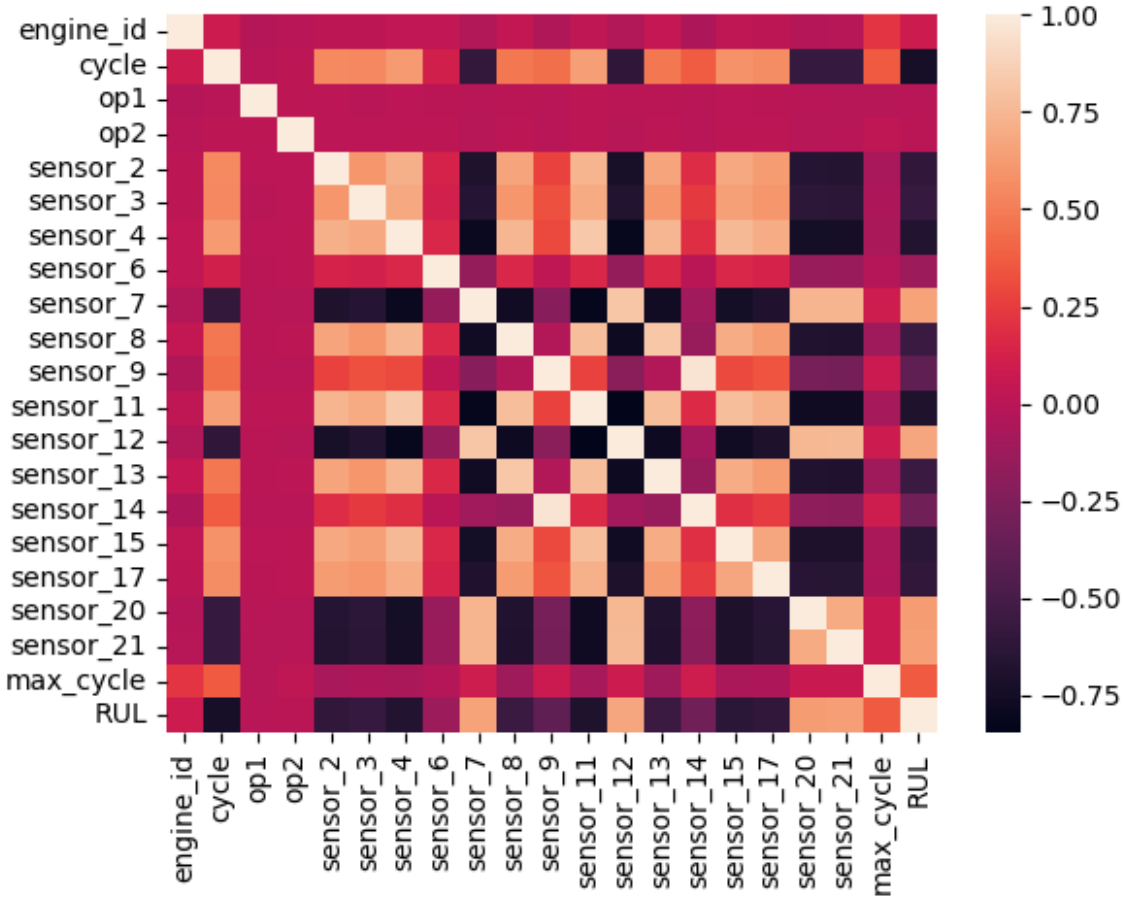
Step 6: Exploratory Data Analysis (EDA)

```
train.hist(figsize=(15,12))
plt.show()
```



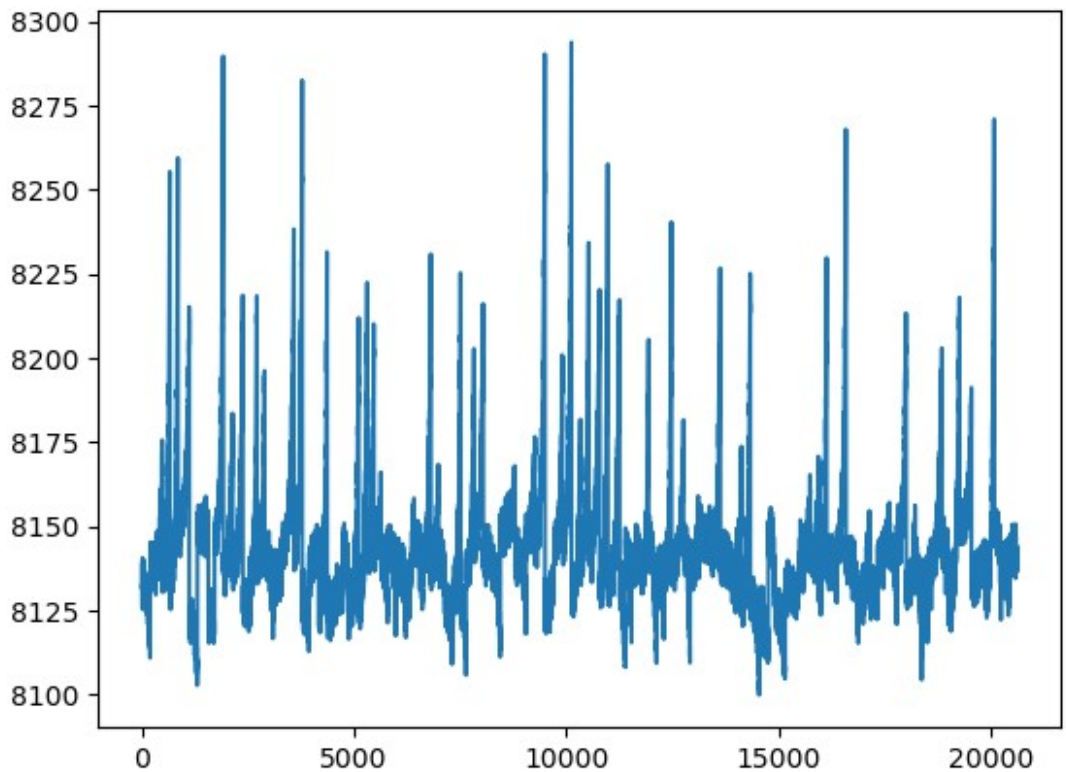

```
corr_matrix = train.corr()

sns.heatmap(corr_matrix)
```

```
<Axes: >
```




```
plt.plot(train['sensor_14'])
```

```
[<matplotlib.lines.Line2D at 0x24448039a60>]
```




```
#Step 7: Feature Selection
train_filtered = train.drop(
['op2','op1','sensor_6','sensor_9'], axis=1)

train_filtered
```

```
       engine_id  cycle  sensor_2  sensor_3  sensor_4  sensor_7
sensor_8  \
0              1      1    641.82   1589.70   1400.60    554.36
2388.06
1              1      2    642.15   1591.82   1403.14    553.75
2388.04
2              1      3    642.35   1587.99   1404.20    554.26
2388.08
3              1      4    642.35   1582.79   1401.87    554.45
2388.11
4              1      5    642.37   1582.85   1406.22    554.00
2388.06
...          ...    ...       ...       ...       ...       ...
...
20626        100    196    643.49   1597.98   1428.63    551.43
2388.19
20627        100    197    643.54   1604.50   1433.58    550.86
2388.23
20628        100    198    643.42   1602.46   1428.18    550.94
2388.24
```

```
20629        100    199    643.23   1605.26   1426.53    550.68
2388.25
20630        100    200    643.85   1600.38   1432.14    550.79
2388.26

       sensor_11  sensor_12  sensor_13  sensor_14  sensor_15
sensor_17  \
0          47.47     521.66    2388.02    8138.62     8.4195
392
1          47.49     522.28    2388.07    8131.49     8.4318
392
2          47.27     522.42    2388.03    8133.23     8.4178
390
3          47.13     522.86    2388.08    8133.83     8.3682
392
4          47.28     522.19    2388.04    8133.80     8.4294
393
...          ...        ...        ...        ...        ...         ..
.
20626      48.07     519.49    2388.26    8137.60     8.4956
397
20627      48.04     519.68    2388.22    8136.50     8.5139
395
20628      48.09     520.01    2388.24    8141.05     8.5646
398
20629      48.39     519.67    2388.23    8139.29     8.5389
395
20630      48.20     519.30    2388.26    8137.33     8.5036
396

       sensor_20  sensor_21  max_cycle  RUL
0          39.06    23.4190        192  191
1          39.00    23.4236        192  190
2          38.95    23.3442        192  189
3          38.88    23.3739        192  188
4          38.90    23.4044        192  187
...          ...        ...        ...  ...
20626      38.49    22.9735        200    4
20627      38.30    23.1594        200    3
20628      38.44    22.9333        200    2
20629      38.29    23.0640        200    1
20630      38.37    23.0522        200    0

[20631 rows x 17 columns]
```

```
train_filtered.shape
```

```
(20631, 17)
```

```
scaler = MinMaxScaler()

#features = train_filtered.drop(['engine_id','RUL'], axis=1)
features = train_filtered.drop(['engine_id','RUL','max_cycle'],
axis=1)

X_scaled = scaler.fit_transform(features)

y = train_filtered['RUL']

X_scaled
```

```
array([[0.        , 0.18373494, 0.40680183, ..., 0.33333333,
0.71317829,
        0.7246617 ],
       [0.00277008, 0.28313253, 0.4530194 , ..., 0.33333333,
0.66666667,
        0.73101353],
       [0.00554017, 0.34337349, 0.36952256, ..., 0.16666667,
0.62790698,
        0.62137531],
       ...,
       [0.54570637, 0.66566265, 0.68497929, ..., 0.83333333,
0.23255814,
        0.05399061],
       [0.54847645, 0.60843373, 0.74602136, ..., 0.58333333,
0.11627907,
        0.23446562],
       [0.55124654, 0.79518072, 0.63963375, ..., 0.66666667,
0.17829457,
        0.21817178]])
```

```
y
```

```
0        191
1        190
2        189
3        188
4        187
        ...
20626      4
20627      3
20628      2
20629      1
20630      0
Name: RUL, Length: 20631, dtype: int64
```

```
#10. Train-Test Split
X_train, X_test, y_train, y_test = train_test_split(
    X_scaled, y, test_size=0.2, random_state=42
)
```

```
X_train.shape
```

```
(16504, 14)
```

```
X_test.shape
```

```
(4127, 14)
```

```
y_test.shape
```

```
(4127,)
```

```
y_train.shape
```

```
(16504,)
```

```
#11. Train Machine Learning Models
rf_model = RandomForestRegressor(random_state=42)
```

```
rf_model.fit(X_train, y_train)
```

```
RandomForestRegressor(random_state=42)
```

```
rf_preds = rf_model.predict(X_test)
```

```
rf_preds
```

```
array([112.62,   9.31, 202.45, ..., 125.19, 130.75, 104.38])
```

```
gb_model = GradientBoostingRegressor(random_state=42)
```

```
gb_model.fit(X_train, y_train)
```

```
GradientBoostingRegressor(random_state=42)
```

```
gb_preds = gb_model.predict(X_test)
```

```
gb_preds
```

```
array([104.28379235,   9.31472409, 199.56790365, ..., 120.12959988,
       121.49822157,  98.48650783])
```

```
from xgboost import XGBRegressor
```

```
xgb_model = XGBRegressor()
```

```
xgb_model.fit(X_train, y_train)
```

```
XGBRegressor(base_score=None, booster=None, callbacks=None,
             colsample_bylevel=None, colsample_bynode=None,
             colsample_bytree=None, device=None,
early_stopping_rounds=None,
             enable_categorical=False, eval_metric=None,
feature_types=None,
```

```
             feature_weights=None, gamma=None, grow_policy=None,
             importance_type=None, interaction_constraints=None,
             learning_rate=None, max_bin=None, max_cat_threshold=None,
             max_cat_to_onehot=None, max_delta_step=None,
max_depth=None,
             max_leaves=None, min_child_weight=None, missing=nan,
             monotone_constraints=None, multi_strategy=None,
n_estimators=None,
             n_jobs=None, num_parallel_tree=None, ...)
```

```
xgb_preds = xgb_model.predict(X_test)
xgb_preds
```

```
array([121.0247   ,   6.2211876, 192.2365   , ..., 118.34315  ,
       118.21848  , 113.39068  ], dtype=float32)
```

```
lr_model = LinearRegression()
```

```
lr_model.fit(X_train, y_train)
```

```
LinearRegression()
```

```
lr_preds = lr_model.predict(X_test)
```

```
lr_preds
```

```
array([ 68.73100422,  29.51563494, 189.57980553, ..., 125.71401025,
       130.83952162, 109.37948762])
```

```
#12. Model Evaluation
def evaluate_model(true, pred):

    mae = mean_absolute_error(true, pred)
    rmse = np.sqrt(mean_squared_error(true, pred))

    print("MAE:", mae)
    print("RMSE:", rmse)

print("Random Forest Performance")
evaluate_model(y_test, rf_preds)

print("Gradient Boosting Performance")
evaluate_model(y_test, gb_preds)

print("XGBRegressor Performance")
evaluate_model(y_test, xgb_preds)

print("Linear Regression Performance")
evaluate_model(y_test, lr_preds)
```

```
Random Forest Performance
MAE: 25.35127695662709
```

```
RMSE: 35.9108478608862
Gradient Boosting Performance
MAE: 25.503620917370533
RMSE: 35.61933880400277
XGBRegressor Performance
MAE: 26.873233168720585
RMSE: 37.82648239860859
Linear Regression Performance
MAE: 30.50926181921916
RMSE: 39.67351685919348
```

```
test_data = pd.read_csv("test_FD001.txt", sep=r"\s+", header=None)
test_data.columns = columns
test_data.head()
```

```
   engine_id  cycle     op1     op2    op3  sensor_1  sensor_2
sensor_3  \
0          1      1  0.0023  0.0003  100.0    518.67    643.02
1585.29
1          1      2 -0.0027 -0.0003  100.0    518.67    641.71
1588.45
2          1      3  0.0003  0.0001  100.0    518.67    642.46
1586.94
3          1      4  0.0042  0.0000  100.0    518.67    642.44
1584.12
4          1      5  0.0014  0.0000  100.0    518.67    642.51
1587.19

   sensor_4  sensor_5  ...  sensor_12  sensor_13  sensor_14  sensor_15
\
0   1398.21     14.62  ...     521.72    2388.03    8125.55     8.4052

1   1395.42     14.62  ...     522.16    2388.06    8139.62     8.3803

2   1401.34     14.62  ...     521.97    2388.03    8130.10     8.4441

3   1406.42     14.62  ...     521.38    2388.05    8132.90     8.3917

4   1401.92     14.62  ...     522.15    2388.03    8129.54     8.4031


   sensor_16  sensor_17  sensor_18  sensor_19  sensor_20  sensor_21
0       0.03        392       2388      100.0      38.86    23.3735
1       0.03        393       2388      100.0      39.02    23.3916
2       0.03        393       2388      100.0      39.08    23.4166
3       0.03        391       2388      100.0      39.00    23.3737
4       0.03        390       2388      100.0      38.99    23.4130
```

```
[5 rows x 26 columns]
```

```
max_cycle_test = test_data.groupby('engine_id')['cycle'].max()

test_data = test_data.merge(max_cycle_test.to_frame(name='max_cycle'),
on='engine_id')

#14. Preprocess Test Data

test_filtered = test_data.drop(['op1','op2','sensor_6','sensor_9'],
axis=1)

test_features = test_filtered.drop(['engine_id'], axis=1)

test_features = test_features[features.columns]

test_scaled = scaler.transform(test_features)



test_predictions = rf_model.predict(test_scaled)

predictions = pd.DataFrame({
    "engine_id": test_filtered["engine_id"],
    "Predicted_RUL": test_predictions
})

predictions.head()
```

```
   engine_id  Predicted_RUL
0          1         206.48
1          1         205.11
2          1         195.19
3          1         210.42
4          1         217.84
```

```
predictions.shape
```

```
(13096, 2)
```

```
true_rul = pd.read_csv(
    "RUL_FD001.txt",
    header=None,
    names=['True_RUL']
)

true_rul.head()
```

```
   True_RUL
0       112
1        98
```

```
2        69
3        82
4        91
```

```
true_rul.shape
```

```
(100, 1)
```

```
final_predictions = predictions.groupby("engine_id").last()

mae = mean_absolute_error(true_rul['True_RUL'],
final_predictions['Predicted_RUL'])

rmse = np.sqrt(mean_squared_error(true_rul['True_RUL'],
final_predictions['Predicted_RUL']))

print("Final MAE:", mae)
print("Final RMSE:", rmse)
```

```
Final MAE: 19.3695
Final RMSE: 26.18636463887265
```